Certified Analytic Geometry on an Explicit K3 Surface

Quantitative Charts, Branch Transitions, and Computer-Assisted Continuation

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Preprint. All quantitative hypotheses of this paper are certified by machine-checkable artifacts; the verification entry point is a single command, described in Appendix F.

Abstract

We equip an explicit algebraic K3 surface with a finite holomorphic atlas whose chart domains, transition maps and branch continuations are certified rather than asserted. The surface is the smooth complete intersection of three diagonal quadrics in $\mathbf{P}^5$ with integer coefficients. Exploiting the diagonal quadratic structure – for which the Taylor remainder is a single explicit term and the relevant Hessian is constant – we prove a uniform quantitative chart lemma with an explicit guaranteed radius, by a self-contained Newton contraction whose uniqueness step is an exact identity rather than an estimate. Sixty explicitly specified chart types cover the surface; on every nonempty overlap the transitions are restrictions of exact algebraic coordinate changes satisfying the cocycle law identically, and they carry the branch bookkeeping with them. We identify the chart pivots as the Plücker coordinates of the row space of the Jacobian, which places the combinatorics of chart changes inside $\mathrm{Gr}(3, 6)$, and we determine how a global sign automorphism acts chart by chart – vertically on some types, moving the base point on others. A validated computation layer, exact over $\mathbf{Q}$ and outward-rounded elsewhere, certifies the quantitative hypotheses together with analytic continuations across chart boundaries; there the continued sheet and the conjugated one are shown to differ by an explicit deck transformation. No Ricci-flat or hyperkähler metric is claimed. The result is a reproducible bridge between an explicit projective model of a K3 surface and theorem-grade local analytic data, suitable for subsequent validated geometric analysis.

Contents

1. Introduction

1.1 The gap between explicit algebraic models and usable analytic geometry

A K3 surface given by explicit projective equations is "explicit" only algebraically. The equations alone do not provide: chart domains with controlled radius; a constructive chart rule valid at every point; transition maps that are effectively verified rather than asserted; robust tracking of root branches; or objects directly consumable by rigorous analytic computation. This paper closes that gap for one concrete surface.

None of this is automatic. A qualitative application of the implicit function theorem yields charts but no certified numerical radius for this instance; computer algebra yields exact local data with no analytic control; a numerical atlas yields neither exactness nor guarantees. What is missing in each case is the *quantitative* statement: a domain of definite size on which a named holomorphic map is proved to exist, transitions that are verified rather than assumed, and a branch of every square root that is followed rather than hoped for. No single ingredient below is new; what is new is assembling them into one globally compatible object whose hypotheses are machine-checkable and whose verification a third party can re-run.

1.2 Main result

> **Theorem A (Certified finite holomorphic atlas).** Let $X \subset \mathbf{P}^5$ be the explicit smooth complete intersection of three diagonal quadrics defined in Section 2. Then X carries a finite holomorphic atlas assembled from 60 explicitly specified chart types, every local chart of every type admitting one and the same certified lower bound on its radius $\rho_{\text{unif}} > 0$ ($\rho_{\text{unif}} \geq 2.10 \times 10^{-12}$ in the normalization of Section 3.5; not sharp, strictly positive, reproducible). On every nonempty overlap, the transition maps are restrictions of explicit algebraic coordinate transformations and satisfy the cocycle identities exactly. The resulting glued analytic space is canonically biholomorphic to X.

The atlas is *finite by compactness* and *quantitative by certificate*: the theorem asserts quantitative existence with a guaranteed radius, not an enumerated list of charts. (The distinction matters and is kept throughout; see Section 4.2.)

1.3 Contributions

(A) A completely explicit K3 model and its branched-cover structure (Section 2). **(B)** A uniform quantitative chart lemma giving true chart domains with certified radius, whose proof is self-contained and exact for quadrics (Section 3). **(C)** Global atlas closure with exact transition and cocycle laws and explicit branch bookkeeping (Sections 4-6). **(D)** A reproducible computer-assisted architecture certifying the quantitative hypotheses without turning the theorem into brute-force enumeration (Sections 7-8).

Scope. We do not claim an explicit Ricci-flat K3 metric, nor a global hyperkähler family. Metric computations appear only as downstream validated probes (Section 9) and are clearly separated from the atlas theorem.

2. The Explicit K3 Surface

2.1 A diagonal CI(2,2,2) model

Let $\mu = (\mu_0, \dots, \mu_5) = (1, 2, 3, 5, 7, 11)$ and, for $k = 0, 1, 2$,

$$F_k(z) \;=\; \sum_{j=0}^{5} \mu_j^k \, z_j^2.$$

Explicitly:

$$\begin{aligned} F_0 &= z_0^2 + z_1^2 + z_2^2 + z_3^2 + z_4^2 + z_5^2, \\ F_1 &= z_0^2 + 2z_1^2 + 3z_2^2 + 5z_3^2 + 7z_4^2 + 11z_5^2, \\ F_2 &= z_0^2 + 4z_1^2 + 9z_2^2 + 25z_3^2 + 49z_4^2 + 121z_5^2, \end{aligned}$$

and $X = \{F_0 = F_1 = F_2 = 0\} \subset \mathbf{P}^5$. The reader can copy the three equations onto a sheet of paper; every constant in the paper is derived from these six integers.

2.2 Smoothness and the K3 property

The Jacobian criterion reduces to Vandermonde nonvanishing: for distinct μ_j the 3×3 minors $V_S = \det(\mu_s^k)_{k,s \in S}$ are nonzero for every triple S, and smoothness follows.

> **Lemma 2.1 (at least three nonzero coordinates).** At every point of X, at least three of the six homogeneous coordinates are nonzero.

Proof. Suppose at most two are nonzero, say z_i and z_j. Writing $w = z^2$, the three equations read $W \cdot (w_i, w_j)^{\mathsf{T}} = 0$ with $W = (\mu_i^k, \mu_j^k)_{k=0,1,2}$. Every 2×2 minor of W is a Vandermonde determinant in two distinct μ's, hence nonzero (checked exactly on all 15 pairs), so W has rank 2 and $w_i = w_j = 0$. Then all coordinates vanish, which is not a point of $\mathbf{P}^5$. $\square$

Smoothness follows at once: for a triple S of nonzero coordinates the corresponding Jacobian minor is the algebraic pivot $\det M_S^{\mathrm{alg}} = 8 V_S z_{s_1} z_{s_2} z_{s_3}$ (Section 3.5), which is nonzero, so the Jacobian has rank 3 everywhere and X is a smooth surface. Adjunction gives $K_X \simeq \mathcal{O}_X$, and the Lefschetz hyperplane theorem gives $H^1(X, \mathbf{Z}) = 0$, hence $h^{1,0} = h^{0,1} = 0$; so X is K3. This part is entirely classical and involves no computation. Lemma 2.1 will be used a second time, for a different purpose, in Section 4.1.

2.3 Projection and branched-cover structure

Choosing three base coordinates and solving the three quadratic equations for the squares of the other three exhibits, on each elimination patch,

$$z_{s_i}^2 = R_i(t_1, t_2, t_3), \qquad i = 1, 2, 3,$$

with R_i explicit rational quadratic expressions (Cramer over the Vandermonde pivot). This gives branched projections $\pi_S : X \dashrightarrow \mathbf{P}^2$ with generic fibre the 2^3 sign choices, a natural sheet action of $(\mathbf{Z}/2)^3$ per patch, and the global diagonal sign action on X itself. Branch divisors are cut by $z_s = 0$.

For the distinguished patch $S = \{3,4,5\}$, $T = \{0,1,2\}$, gauge $z_0 = 1$ and base $(u,v) = (z_1, z_2)$, Cramer over the Vandermonde pivot ($|V_S| = 48$) gives the three radicands exactly:

$$R_3 = -5 - \tfrac{15}{4}u^2 - \tfrac{8}{3}v^2, \qquad R_4 = 5 + \tfrac{27}{8}u^2 + 2v^2, \qquad R_5 = -1 - \tfrac{5}{8}u^2 - \tfrac{1}{3}v^2.$$

A remark that shapes the whole paper: since $F_0 = \sum_j z_j^2$ is positive definite, $X(\mathbf{R}) = \varnothing$ – the surface has no real points at all for this μ. Accordingly the three radicands above have *constant sign* on the real (u,v) slice, and the branch loci are invisible there. The regime structure that the continuation machinery of Section 7 navigates lives in the complex directions: in the slice $v = 0$ of the complex u-plane the loci $\operatorname{Re} R_s = 0$ are three nested rectangular hyperbolas $x^2 - y^2 = -a_s/b_s$, and crossing one flips exactly one solved line between the principal and the canonical determination (Figure 1).

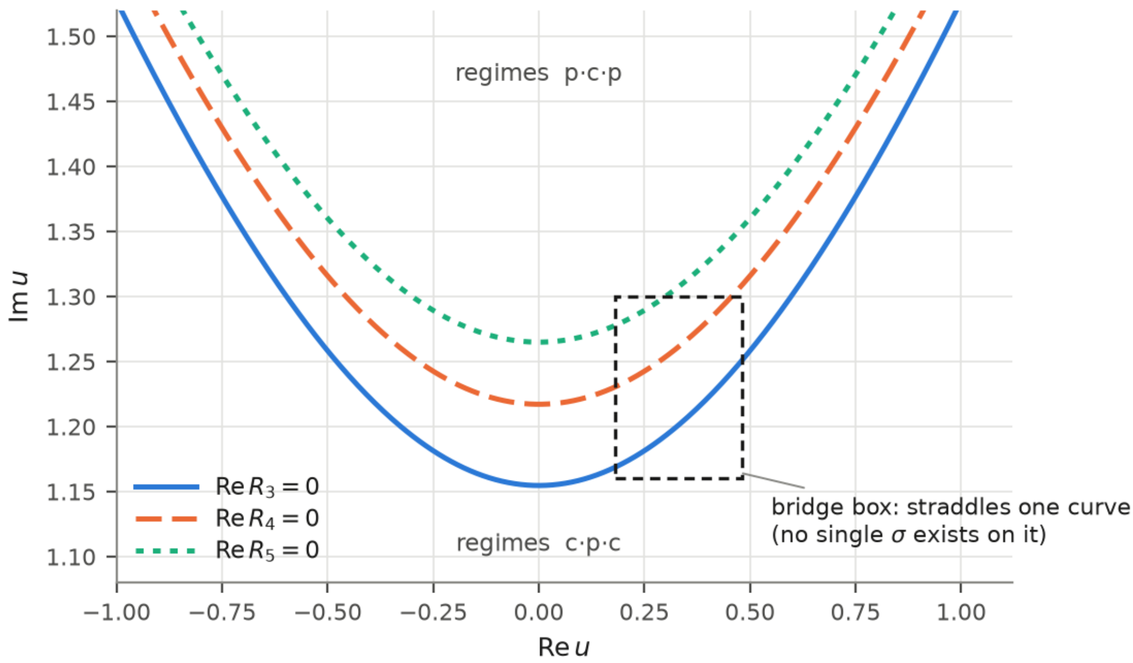

Figure 1: The three loci $\operatorname{Re} R_s = 0$ of the chart $S = \{3,4,5\}$, $z_0 = 1$ in the complex base slice $v = 0$: nested rectangular hyperbolas whose rational coefficients are derived from μ, the regime pattern on either side, and a bridge box straddling one curve. *(data)*

The deck symmetries are the diagonal sign matrices $\sigma_j = \operatorname{diag}(1, \dots, -1, \dots, 1)$ (the -1 in slot j): each preserves every F_k, since the coordinates enter only through their squares. Modulo the projective scalar -1 they generate a group of order 32, and $\sigma_1, \dots, \sigma_5$ suffice, because $\sigma_0 = -\sigma_1\sigma_2\sigma_3\sigma_4\sigma_5$ in $\mathbf{P}^5$.

The branch locus of π_S is $\{R_{s_1} R_{s_2} R_{s_3} = 0\}$. For the distinguished patch this discriminant is the explicit degree-six polynomial

$$\tfrac{2025}{256}u^6 + \tfrac{465}{32}u^4v^2 + \tfrac{2235}{64}u^4 + \tfrac{53}{6}u^2v^4 + \tfrac{1031}{24}u^2v^2 + \tfrac{205}{4}u^2 + \tfrac{16}{9}v^6 + \tfrac{118}{9}v^4 + \tfrac{95}{3}v^2 + 25.$$

This explains *before any computation* why sign patterns exist and why every chart carries branch bookkeeping.

2.4 Chart notation

A chart type is a pair (S, g): a triple $S \subset \{0, \ldots, 5\}$ of solved coordinates (20 triples) and a gauge coordinate $g \notin S$ normalized to $z_g = 1$ (3 choices per triple after the selector; 60 types in total). Solved variables $w = z_S \in \mathbf{C}^3$; base variables $b = (u, v) \in \mathbf{C}^2$; sheet sign pattern $\varepsilon \in \{\pm 1\}^3$. Two layers must be kept apart, and the paper keeps them apart throughout. The **algebraic layer** uses the equations as written: $V_S \in M_3(\mathbf{Z})$ and every identity of Sections 2, 4 and 5 lives over $\mathbf{Q}$. The **normalized layer** rescales rows, $\tilde{F}_k = F_k/c_k$ with $c_k^2 = \sum_j \mu_j^{2k}$; this is a conditioning convention for the certificate, not part of the geometry. The *squares* c_k^2 are integers; the c_k themselves are irrational, and the certificate handles them as algebraic numbers by outward-rounded rational bracketing. *Worked example.* Take $S = \{3, 4, 5\}$ and $g = 0$: the solved variables are $w = (z_3, z_4, z_5)$, the gauge is $z_0 = 1$, the base is $(u, v) = (z_1, z_2)$, and a point of the chart is $z_s = \varepsilon_s \sqrt{R_s(u, v)}$ for the radicands of Section 2.3 and a sign pattern $\varepsilon \in \{\pm 1\}^3$. The normalization constants have exact integer squares, $c_0^2 = 6$, $c_1^2 = 209$, $c_2^2 = 17765$ (the c_k themselves are irrational – see the two layers above), and the Vandermonde pivot of this triple is $|V_S| = 48$. Every constant appearing later in the certificate for this chart is a rational function of these.

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3. Quantitative Local Charts

3.1 Local elimination

For an admissible type (S, g) write $G(w; b) := \tilde{F}(w, b, 1)$. Two Jacobians must be kept apart, and confusing them is easy because they differ only by the row normalization of Section 2.4:

$$M_S^{\mathrm{alg}} := \partial_w F, \qquad \tilde{M}_S := \partial_w \tilde{F} = \partial_w G, \qquad \det M_S^{\mathrm{alg}} = c_0 c_1 c_2 \cdot \det \tilde{M}_S.$$

Admissibility is the algebraic pivot condition

$$q_S(Z) \ := \ \frac{|\det M_S^{\mathrm{alg}}(Z)|}{\|Z\|^3} \ > \ m_{\mathrm{alg}}, \qquad m_{\mathrm{alg}} = 4$$

(m_{alg} preregistered, below the observed floor 4.8), stated once and for all on the **projective invariant** q_S – equivalently, on the unit-norm representative $\|Z\| = 1$, which is the slice the coverage certificate enumerates. Since $\det M_S^{\mathrm{alg}}$ is homogeneous of degree 3 (Section 3.5), the bare inequality $|\det M_S^{\mathrm{alg}}| > 4$ would depend on the choice of representative, whereas the vanishing condition $\det M_S^{\mathrm{alg}} \neq 0$ of Lemma 4.1 does not, and is used as such. Throughout, U_S denotes the open $\{q_S > m_{\mathrm{alg}}\}$. It is the algebraic pivot that enters the coverage argument of Section 4.1 and the coordinate floor of Lemma 3.1, both of which use the integer factorization $\det M_S^{\mathrm{alg}} = 8 V_S z_{s_1} z_{s_2} z_{s_3}$ with $V_S \in \mathbf{Z}$. The **Newton estimates of Section 3.4 use the row-normalized** $\tilde{M}_S$, whose smallest singular value is the σ of Proposition 3.2. The same symbol is never used for both.

3.2 Why the present model is unusually tractable

For diagonal quadrics the following identities are *exact*, not estimates:

- (E1) $\partial_w G(w;b) = \tilde{M}(w) = 2\tilde{V}_S \operatorname{diag}(w)$, independent of b;
- (E2) $G(w;b) - G(w_0;b) - \tilde{M}(w_0)(w-w_0) = \tilde{V}_S \cdot ((w-w_0)^2)$ (componentwise square): the Taylor remainder is a single explicit quadratic term, the "Hessian" is constant;
- (E3) $G(w_0;b) - G(w_0;b_0) = \tilde{V}_{uv} \cdot (b^2 - b_0^2)$.

No numerical approximation of any higher derivative is ever needed. This is the structural reason the algebraic layer closes over $\mathbf{Q}$, the row normalization being the only place where algebraic numbers enter – and there they are bracketed, never approximated.

3.3 The pivot-to-coordinate floor

The condition bounds a determinant; the Newton estimates need a bound on the *coordinates*. Bridging the two is a short argument that will be used twice – here for σ, and in Section 4.3 for the distinctness of sheet sign patterns. Sheets of π_S do merge, exactly where a solved coordinate vanishes, so a uniform bound cannot hold on all of X and has to be earned on the domains actually used.

Lemma 3.1 (pivot-to-coordinate floor). Let a_s, b_s, c_s be the rational coefficients of the radicand $R_s = a_s + b_s u^2 + c_s v^2$ and set $B_s := (|a_s| + |b_s| + |c_s|)^{1/2}$. On a chart domain $U_S \cap C_{S,g}$ every solved coordinate is bounded below:

$$|z_{s_i}| \ > \ \max\left\{ \frac{m_{\mathrm{alg}}}{8\,|V_S| \prod_{j\neq i} B_{s_j}}, \ \ \left(|a_{s_i}| - |b_{s_i}| - |c_{s_i}|\right)^{1/2} \right\} \ > \ 0,$$

the second entry being used only when the quantity under the root is positive. Over the 60 chart types the worst such bound is 9.5711×10^{-4} (the certified value $9.571170\ldots \times 10^{-4}$ rounded outward), attained at $S = \{3,4,5\}$ in the gauges z_1 and z_2 alike, at the solved coordinate z_5; in the gauge z_0 the same type has floor 2.203×10^{-3}, at z_4. Consequently two sign patterns differing in slot s give points at distance at least twice that in the s-th coordinate: the eight sheets are uniformly separated on certified domains.

Proof. First the threshold has to be turned into a statement about the representative actually used. In the gauge $z_g = 1$ we have $\|Z\| \geq 1$, so $q_S(Z) > m_{\mathrm{alg}}$ gives $|\det M_S^{\mathrm{alg}}(Z)| = q_S(Z)\,\|Z\|^3 > m_{\mathrm{alg}}$: the projective condition implies the plain inequality on the gauge representative, which is the form the factorization consumes. From there, the pivot threshold bounds a **product**, not its factors: by the factorization of Section 3.5, $|\det M_S^{\mathrm{alg}}| = 8|V_S| \prod_i |z_{s_i}| > m_{\mathrm{alg}}$ gives $\prod_i |z_{s_i}| > m_{\mathrm{alg}}/(8|V_S|)$. The sector selector supplies the missing upper bounds: $|u|, |v| \leq 1$, so $|z_s|^2 = |R_s(u,v)| \leq |a_s| + |b_s| + |c_s| = B_s^2$. Dividing the product bound by the two upper bounds B_{s_j}, $j \neq i$, gives the first entry. The second is independent of the condition: the same triangle inequality read from below gives $|R_{s_i}| \geq |a_{s_i}| - |b_{s_i}| - |c_{s_i}|$, which is the sharper bound on the lines where the constant term dominates. All constants are exact rationals with outward-rounded roots, and the worst case is computed over all 60 types, gauge by gauge. The two entries behave differently under a change of gauge. Changing g within T permutes the columns of w_T and leaves each row sum $|a_s| + |b_s| + |c_s|$ unchanged, so B_s, and with it the first entry, is gauge-independent. The second entry is not: which coefficient plays the constant term a_s depends on which coordinate is set to 1. For $S = \{3, 4, 5\}$ the first entry alone gives 9.5712×10^{-4} at z_5; in the gauge z_0 the second entry lifts z_5 to about 0.204 and the floor of the type becomes 2.203×10^{-3}, at z_4, while in the gauges z_1 and z_2 the second entry does not apply and the floor stays at 9.5712×10^{-4}. The certificate records the three gauges of every triple separately, and the worst case over the 60 types is attained at $S = \{3, 4, 5\}$, $g \in \{1, 2\}$. □

So the pivot threshold is precisely the guard that keeps every chart away from its own branch locus.

3.4 A quantitative inverse-function lemma

Proposition 3.2 (Uniform quantitative chart lemma). Fix a type (S, g), a lower bound $\sigma \le s_{\min}(\tilde{M}(w_0))$ – supplied by $s_{\min}(\tilde{M}(w_0)) \ge 2\, s_{\min}(\tilde{V}_S) \cdot \min_{s \in S} |z_s|$ together with Lemma 3.1 – and the exact quadratic remainder bound (E2) with constant $L = 2\,\sigma_{\max}(\tilde{V}_S)$. For every admissible centre the simplified Newton iteration for $G(\cdot\,; b) = 0$ contracts on a ball of explicit radius: with residual $\eta \le a_{S,g}\, \rho(2 + \rho)/\sigma$, where $a_{S,g}$ bounds the norm of the base-variable block $\tilde{V}_{uv}$ of the normalized Vandermonde and $|b_0| \le 1$ by the sector selector of Section 3.5, and $h = L\eta/\sigma < 1/2$, there is a unique zero at distance $r^* = (1 - \sqrt{1 - 2h}\,)\, \sigma/L \le 2\eta$, unique in the open ball $\|w - w_0\| < \sigma/L$ by the midpoint identity $0 = \tilde{M}((w + w')/2)(w - w')$, which is exact for quadrics. The local solution is holomorphic in b: with the centre held fixed and G quadratic, every simplified-Newton iterate is a polynomial in the base variables b, and the limit is uniform. All hypotheses are verified by exact rational checks on all 60 types.

The proof is self-contained (no external quotable constant is load-bearing) and is given in Appendix B.

The floor of Lemma 3.1 and the displacement bound of Proposition 3.2 close, together, the step that obligation **O5** of Section 4.3 consumes:

Corollary 3.3 (charts avoid their own branch locus). On a chart domain the solved coordinates stay bounded away from zero *throughout the certified ball*, not merely at its centre. Indeed the branch locus of π_S is $\{R_{s_1} R_{s_2} R_{s_3} = 0\}$, that is, the vanishing of a solved coordinate; Lemma 3.1 bounds each one below by $\underline{z}_{S,g}$ at the centre, and Proposition 3.2 confines the solution to $\|w - w_0\| \le r^* \le 2\eta$, so every point of the chart satisfies $|z_{s_i}| \ge \underline{z}_{S,g} - 2\eta$. Over the 60 types the ratio $2\eta/\underline{z}_{S,g}$ is at most 1.72×10^{-2}, attained at $S = \{0, 3, 5\}$, $g = 1$ ($2\eta = 2.22 \times 10^{-5}$ against a floor of 1.29×10^{-3}): the displacement is two orders of magnitude below the floor it would have to cross. Hence every certified chart domain lies in X° for its own projection, the eight signed sheets are distinct on it, and the sign pattern is unambiguous.

The constants η and $\underline{z}_{S,g}$ are both serialized per type in the chart certificate, so the ratio above is read off the artifact rather than recomputed by hand.

3.5 Projective normalization

The radius must not depend on a naive choice of projective representative. The gauge is fixed by $z_g = 1$ on the affine open $A_g = \{z_g \neq 0\}$, and centres are selected in the closed sector $C_{S,g} = \{|z_g| = \max_t |z_t|\}$ (selector only; it guarantees $|u|, |v| \leq 1$ and is never used as an open cover). The homogeneity is exactly right for this. Since $\partial \tilde{F}_k / \partial z_s = 2(\mu_s^k / c_k) z_s$, we have $\tilde{M}_S = 2\tilde{V}_S \, \mathrm{diag}(z_S)$ and therefore the exact factorization

$$\det M_S^{\mathrm{alg}} \;=\; 8\, V_S\, z_{s_1} z_{s_2} z_{s_3}, \qquad V_S \in \mathbf{Z},$$

for the unnormalized equations – an identity entirely over $\mathbf{Z}$ – and $\det \tilde{M}_S = \det M_S^{\mathrm{alg}} / (c_0 c_1 c_2)$ after row normalization. Both were verified symbolically on all 20 triples. It is homogeneous of degree 3 in Z, so

$$q_S(Z) \;=\; \frac{|\det M_S^{\mathrm{alg}}(Z)|}{\|Z\|^3}$$

– the quantity on which Section 3.1 states the admissibility condition – is invariant under $Z \mapsto \lambda Z$: the certified radius is read off a genuine projective invariant, not off a choice of representative. (Replacing the exponent 3 by 2 destroys the invariance – this is one of the negative controls.) The sector $|z_g| = \max_t |z_t|$ selects a representative; it is never used as an open set of a cover.

3.6 The certified radius

Quantity	Exact / certified bound
algebraic pivot threshold m_{alg} (preregistered)	4 (observed floor 4.8)
chart types	60 (20 triples $\times$ selector)
$\sigma_{\min}(\tilde{V}_S)$	$\geq |\det \tilde{V}_S| / \|\tilde{V}_S\|_F^2$ (exact rational)
Lipschitz constant L	$\leq 2\,\sigma_{\max}(\tilde{V}_S) \leq 2\|\tilde{V}_S\|_F$
contraction certificate	$h < 1/2$ on all 60 types
per-type radius $\rho_{S,g}$	$\in [2.10 \times 10^{-12},\ 1.71 \times 10^{-4}]$ (outward-rounded)
guaranteed uniform radius $\rho_{\mathrm{unif}} := \min_{S,g} \rho_{S,g}$	$\geq 2.10 \times 10^{-12}$

The uniform radius is a Frobenius/determinant bound: strictly positive and reproducible, deliberately not sharp. A uniform radius is a single number, not an interval: the *per-type* radii $\rho_{S,g}$ range over eight decades (Figure 2) and ρ_{unif} is their minimum. The per-type values are in the certificate artifact.

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4. From Local Charts to a Global Atlas

4.1 Global admissibility

Every point of X admits at least one admissible triple: the pivot opens $U_S = \{q_S > m_{\mathrm{alg}}\}$ cover X (coverage margin $\tau = 0.6$; the zero set of all pivots is empty). Coverage is *qualitatively* a two-line consequence of

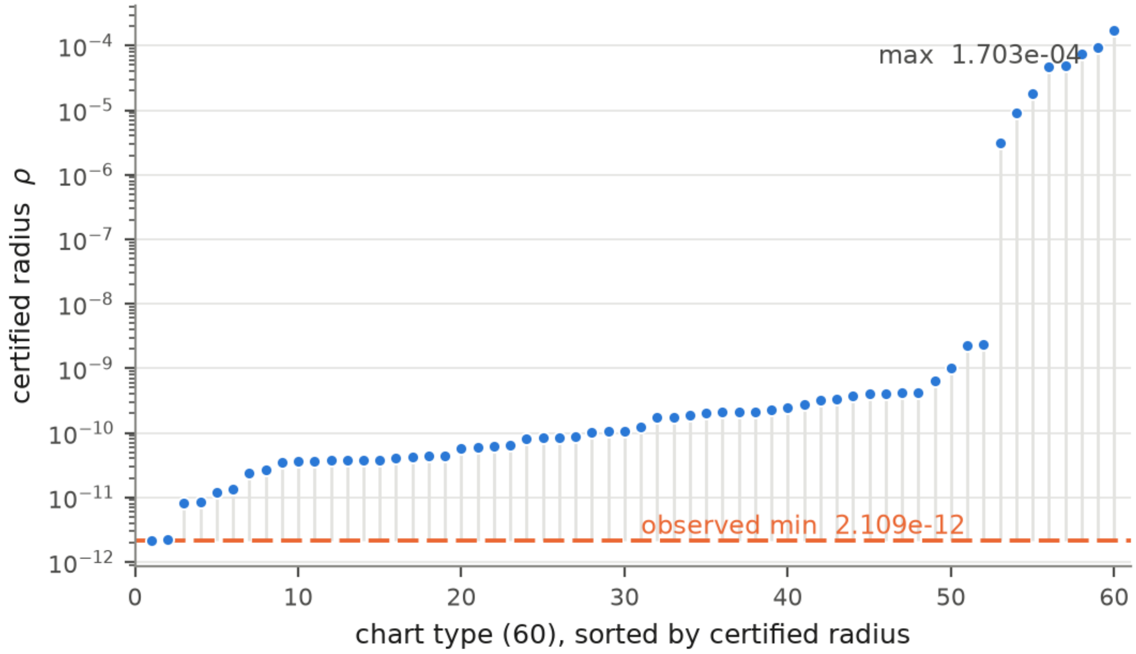

Figure 2: Certified radius across the 60 chart types on a log scale, spanning $[2.10 \times 10^{-12}, 1.71 \times 10^{-4}]$ (endpoints rounded outward). The dashed line is the observed minimum over the types, 2.109×10^{-12}; the guaranteed floor quoted in the text, $\rho_{\text{unif}} \geq 2.10 \times 10^{-12}$, is that value rounded outward. Regenerated from the repaired chart certificate. *(data)*

Lemma 2.1. By the factorization of Section 3.5, $\det M_S^{\text{alg}} \neq 0$ if and only if the three solved coordinates are all nonzero; Lemma 2.1 provides such a triple at every point; hence

> **Lemma 4.1 (coverage).** The nonvanishing pivot opens $W_S := \{\det M_S^{\text{alg}} \neq 0\}$, S ranging over the 20 triples, cover X. (Vanishing is a projective condition, so W_S needs no normalization; the symbol U_S is reserved throughout for the *quantitative* open $\{q_S > m_{\text{alg}}\} \subset W_S$.)

What the large enumeration audit adds is not the covering – it is the *quantitative floor*: a positive threshold m_{alg} (preregistered at $m_{\text{alg}} = 4$, against an observed minor floor 4.8) and a coverage margin $\tau = 0.6$, so that the opens $\{q_S > m_{\text{alg}}\}$ still cover, with room to spare. That audit is an independent certificate, referenced by hash from the chart certificate, and its box enumeration belongs in the supplement.

4.2 Uniformity and finiteness

The chart types form a finite family (60) with uniformly controlled constants; compactness of X extracts a finite subcover of the local chart balls. The theorem is *quantitative existence with guaranteed radius*: no enumerated atlas is claimed, and the certification architecture explicitly forbids the stronger wording (a firewall check turns red if "explicit enumerated atlas" is asserted).

4.3 The atlas theorem, and the glued space as a corollary

The charts of Proposition 3.2 are parametrizations of open subsets of one and the same $X \subset \mathbf{P}^5$, and on every overlap their transitions are the ambient projective coordinate identifications of Section 5. Nothing has to be reconstructed:

Theorem 4.2 (Global certified atlas). The local parametrizations of Proposition 3.2 cover X (Lemma 4.1), and on every nonempty overlap their transition maps agree with the ambient projective coordinate identifications and satisfy the cocycle law exactly (Section 5). They therefore constitute a finite holomorphic atlas on X, all of whose charts carry the uniform certified radius $\rho_{\mathrm{unif}} > 0$.

The gluing statement is then a corollary rather than the mechanism of proof:

Corollary 4.3. The analytic space $\mathcal{X}_{\mathrm{atlas}} = \left(\bigsqcup_\alpha U_\alpha\right)/\sim$ obtained by gluing these parametrizations along the certified transitions is canonically biholomorphic to X:

$$\boxed{\mathcal{X}_{\mathrm{atlas}} \cong_{\mathrm{bihol}} X.}$$

Reading the corollary as a construction from data – hand someone the chart types, the radii and the transition formulas, and they rebuild X – is what makes it worth stating; it is not what carries Theorem 4.2.

For the corollary the obligations are the standard ones, and it is worth recording which are algebra and which consume certified data.

	Obligation	Proof mode	Support
O1	$\sim$ is an equivalence relation	exact algebra	the cocycle of Proposition 5.1, together with $P_{SS} = \mathrm{Id}$ and $P_{S'S}^{-1} = P_{SS'}$
O2	Φ is well defined	exact algebra	every chart section solves $F_0 = F_1 = F_2 = 0$ by construction (elimination), hence lands in X
O3	Φ is a local biholomorphism	certificate	Proposition 3.2: certified radius, contraction $h < 1/2$
O4	Φ is surjective	certificate	the pivot opens cover X (Lemma 4.1)
O5	Φ is injective	exact algebra + sign pattern	on a chart domain the three solved coordinates are bounded away from zero, so the eight signed sheets are *distinct there* and the sign pattern separates them; $\sim$ therefore captures all coincidences of chart points

	Obligation	Proof mode	Support
O6	$\mathcal{X}_{\text{atlas}}$ is second countable and Hausdorff	topology	the atlas is finite by compactness (Section 4.2); a continuous bijection that is a local homeomorphism is open, hence a homeomorphism, so $\mathcal{X}_{\text{atlas}}$ inherits Hausdorffness from X

Obligation **O5** uses the pivot-to-coordinate floor of Lemma 3.1: on a chart domain the three solved coordinates are bounded away from zero, so the eight signed sheets are distinct there. The sheets that merge over B are not lost – they are covered by charts of a different triple, which Lemma 2.1 always provides.

Three obligations are exact algebra, two are certificates, one is topological. Note that Hausdorffness is *not* bought with a separation estimate: once Φ is a bijective local homeomorphism it is open, and the quotient inherits the topology of X. Lemma 3.1 is therefore not needed for O6 – it is a quantitative lemma in its own right, used for the sign pattern distinctness of **O5** and for the lower bound on σ in Section 3.

4.4 Topology as an audit, not the proof

Homology of the nerve is used only as an independent consistency control. Homology matching is *not* used to identify the glued analytic space with X; the identification is the canonical map Φ of Theorem 4.2.

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5. Exact Transition Maps

5.1 Algebraic chart changes

Between two types (S, g) and (T, h) the transition is the restriction of an exact algebraic coordinate change: a permutation of coordinates, a projective rescaling by z_h/z_g, and a re-selection of solved roots. Two closed formulas make this concrete. An **elementary swap** $S = \{3,4,5\} \to S' = \{2,4,5\}$ acts on the squares by

$$w_2 = -6w_3 - 15w_4 - 45w_5, \qquad w_4 = w_4, \qquad w_5 = w_5,$$

that is, by a matrix whose last two rows are those of the identity: exactly one row changes, the one carrying the newly solved coordinate. On the coordinates themselves the transition is $z_{s'} = \varepsilon_{s'}\sqrt{(P_{S'S}w_S)_{s'}}$, so the sheet sign pattern ε is part of the data of the map – this is the branch-crossing content, and it is why transitions are recorded together with their sign patterns. A **gauge change** $(S, g) \to (S, g')$ is the projective rescaling by $z_g/z_{g'}$ and touches no sign pattern.

5.2 Exact cocycle law

Underlying the chart transitions there is a purely algebraic cocycle, at the level of the squares. Since the quadrics are linear in $w_j = z_j^2$, the constraint $V_{\text{full}}\, w = 0$ determines all six squares from any admissible triple; hence for every ordered pair (S, S') there is an exact rational 3×3 matrix $P_{S'S}$ with $w_{S'} = P_{S'S}\, w_S$ on X. These satisfy $P_{SS} = \text{Id}$ (20/20), $P_{SS'}P_{S'S} = \text{Id}$ (400/400) and

$$P_{S''S'}\, P_{S'S} \;=\; P_{S''S} \qquad \text{on all } 20^3 = 8000 \text{ ordered triples, in exact rational arithmetic.}$$

On every triple overlap of charts, $\Phi_{\gamma\beta} \circ \Phi_{\beta\alpha} = \Phi_{\gamma\alpha}$ *exactly* – these are algebraic identities in exact arithmetic, not numerical statements with tolerance. The certification layer verified the full transfer/cocycle panel (3540/3540 identities). The paper carefully distinguishes exact algebraic identity from certified numerical applicability on a given domain.

5.3 Finite transition generators

The twenty solved triples are not a combinatorial device imported from outside: they are the standard coordinate charts of a Grassmannian. The Jacobian $dF = 2\,V \operatorname{diag}(z)$ is a 3×6 matrix, so its row space is a point of $\text{Gr}(3, 6)$, and its S-minor is exactly the pivot

$$\det M_S^{\text{alg}} \;=\; 8\, V_S\, z_{s_1} z_{s_2} z_{s_3}$$

(verified on all 20 minors). **The pivots are the Plücker coordinates of the row space of dF.** The solved triples are therefore the standard coordinate charts of $\text{Gr}(3, 6)$, pulled back along $p \mapsto \text{rowspace}\, dF(p)$, and an elementary swap $|S \cap S'| = 2$ is precisely the standard chart change there; the Johnson graph $J(6, 3)$ is the adjacency graph of those elementary chart changes. (It is not the nerve of the cover: any two standard charts of $\text{Gr}(3, 6)$ meet, so that nerve would be the full simplex on 20 vertices.)

With that reading, we exhibit a small generating set. Two elementary moves act on chart types (S, g):

- an **elementary swap** $S \to S'$ with $|S \cap S'| = 2$ (one solved coordinate is exchanged against one base coordinate);
- a **gauge change** $(S, g) \to (S, g')$ with $g' \in T \setminus \{g\}$ (projective rescaling by $z_g/z_{g'}$: the source gauge has $z_g = 1$, so reaching $z_{g'} = 1$ divides the representative by $z_{g'}$).

Together with the sheet flips of Section 6.2 these generate everything, with an explicit bound on word length:

> **Proposition 5.1 (finite generation with length bound).** The 20 solved triples with adjacency $|S \cap S'| = 2$ form the Johnson graph $J(6, 3)$: 20 nodes, 90 edges, connected, of diameter 3. The 60 chart types under {elementary swap, gauge change} – the swap keeping g fixed, the gauge change keeping S fixed – form a connected graph of diameter 4; the extra step over $J(6, 3)$ is real, a witness being $(\{0, 1, 2\}, 3) \to (\{0, 2, 4\}, 3) \to (\{0, 2, 4\}, 1) \to (\{2, 3, 4\}, 1) \to (\{3, 4, 5\}, 1)$,

where the gauge 3 must be moved before the triple can reach $\{3, 4, 5\}$. Consequently, by the co-cycle of Section 5.2, **every chart transition is a word of length at most** 4 **in elementary swaps and gauge changes, composed with a single sheet-group element.**

Remark. Both moves are necessary. Removing gauge changes breaks the type graph into **six connected components of ten types each** – freezing the gauge g confines S to $J(5, 3)$ on the five remaining coordinates – so no composition of swaps alone realizes a gauge change.

---

6. Branch Geometry

6.1 Sheet labels and the radical description

The three quadrics are *linear in the squares* Z_j^2: solving the Vandermonde system gives, for any two complementary triples S, T,

$$w_S \;=\; -V_S^{-1} V_T\, w_T \qquad (w_S = (z_s^2)_{s\in S}),$$

with exact rational coefficients (the largest coefficient over all triples is $C = \max_S \|V_S^{-1} V_T\|_\infty = 112$, exactly). Each patch therefore carries the eight signed sheets $z_s = \varepsilon_s\sqrt{R_s}$, $\varepsilon \in \{\pm 1\}^3$, and the signed radical descriptions are exhaustive. They are *distinct* exactly off the branch locus: over X° a point determines one sign pattern, whereas along a branch stratum the sign patterns are identified precisely in the slots whose solved coordinate vanishes (there $+\sqrt{0} = -\sqrt{0}$). Every certified chart domain lies in X° for its own projection, by Corollary 3.3, so on chart data the sign pattern is unambiguous. The sheet sign pattern ε is part of the chart data.

A discipline that the certification imposed and the paper keeps: **sign patterns are derived, never defaulted.** In the bridge construction of Section 7 the sign pattern left at the default $(1, 1, 1)$ was *refused* by the exact regluing test, and the derived sign pattern $(1, -1, -1)$ closed it; the refusal was correct behaviour, not an error.

6.2 Deck transformations

Diagonal sign matrices $Z_j \mapsto \pm Z_j$ preserve each quadric $Q_m(Z) = \sum_a \mu_a^m Z_a^2$ trivially (coordinates enter only through their squares), so every such matrix induces a holomorphic automorphism of X. Modulo the projective scalar -1 they form the **global diagonal sign automorphism group**

$$G_{\mathrm{sign}} \simeq (\mathbf{Z}/2)^5, \qquad |G_{\mathrm{sign}}| = 32,$$

which is *not* the deck group of any one projection. For a fixed π_S the deck group is the vertical subgroup $G_{\mathrm{deck}}(S) \simeq (\mathbf{Z}/2)^3$ of order 8 and index 4 (Section 6.3), acting as the sheet permutations over X°. The certification layer works with one distinguished non-scalar representative,

$$D \;=\; \mathrm{diag}(+1,-1,+1,+1,+1,-1),$$

an involution which is *not* a projective scalar, and certifies its exact reuse of chart data on the 64 bridge domains of Section 7.2: on every bridge the identity $Z^{\mathrm{conj}}_{\mathrm{upper}} = D \cdot Z_{\mathrm{bridge}}$ holds at the level of the certified coefficients, and the domains on which it is certified are stratified, 36 of the 64 being open in all four ambient coordinates and 28 relative-open (24 relative to one real face, 4 to two). What is *not* trivial about D – and is measured, not asserted – is Section 7.3: D is the exact discrepancy between conjugation and continuation.

6.3 One abstract transformation, different chart realizations

The same abstract deck element acts differently depending on the chart type. In a chart where a flipped coordinate is a *solved* variable, the element is a sheet permutation over a fixed base point (a vertical arrow in the sign pattern); in a chart where that coordinate is a *base* or *gauge* variable, the same element moves the base point. The certification history contains a sharp instance of the distinction: an audit revealed that 316 atlas nodes had been implicitly typed as 316 vertical-T arrows, which is false – the correct statement types the *carrier* of the transformation chart by chart, and one derived predicate is not universal across chart types. In the paper's language:

$$\text{abstract deck action} \;\neq\; \text{chartwise vertical action},$$

and the atlas bookkeeping keeps the two straight by typing the carrier (S, g) of every arrow. The split is quantitative: the deck group has order 32 (diagonal signs $\{\pm 1\}^6$ modulo the projective scalar), generated by five flips; for each chart type the *vertical* subgroup – the elements fixing the base point, i.e. those whose sign vector is constant on the complement T – has order 8 and **index** 4, the quotient $(\mathbf{Z}/2)^2$ moving the base. For the distinguished element $D = \mathrm{diag}(+1,-1,+1,+1,+1,-1)$ of Section 6.2 this is completely explicit: D is vertical for exactly 4 of the 20 triples, i.e. 12 of the 60 chart types, and moves the base point on the remaining 48 (Figure 3, where the pattern is visible: vertical exactly on the triples containing both coordinates that D flips). Deck statements about *sheets* are made relative to a chart's sign pattern convention (source and target sheets qualified), never as absolute class labels.

6.4 Away from the branch locus

Let $X^\circ = X \setminus \pi^{-1}(B)$ with B the branch divisor of the chosen projection. Only on X° is the projection an etale covering with a genuine local system of sheets; on B the correct language is ramification. All local-system statements in this paper are restricted to X°.

Figure 3: The 20 triples $\times$ 3 gauges, marking where D is vertical (12 types) and where it moves the base (48), exhibiting the pattern: vertical exactly when $S \supseteq \{1,5\}$, the coordinates D flips. *(data)*

7. Computer-Assisted Analytic Continuation

7.1 Why continuation needs certification

A root formula chosen at a centre does not, by itself, stay on the correct branch over a whole domain: the radicand R can cross the branch cut. We fix the convention throughout: $\sqrt{\cdot}$ denotes the principal square root, with cut $(-\infty, 0]$, so the principal determination of $\sqrt{R}$ is available exactly where R avoids $(-\infty, 0]$, and a convexity lemma for the certified boxes turns this into a checkable sign condition. The rotated determination

$$\sqrt{R}_{\mathrm{rot}} \;=\; \sigma\, i\, \sqrt{-R}_{\mathrm{principal}}$$

costs no new guard: it is available exactly where R avoids $[0, +\infty)$, since that holds iff $-R$ avoids $(-\infty, 0]$, so the existing guard applies verbatim to $-R$. Together the two determinations cover $\mathbf{C} \setminus \{0\}$. The sign σ is not free – it is fixed by the argument half-plane: $\sigma = +1$ for $\arg R \in (0, \pi]$, $\sigma = -1$ for $\arg R \in (-\pi, 0)$.

7.2 Certified bridge domains

A *bridge* is a box straddling a regime boundary on which the section is built bilaterally. In the certified panel (64 bridge charts over a witness arc): the analytic regime of each solved line is assigned *by the certified sign of* $\mathrm{Re}\, R$, without trial (pattern (principal, canonical, canonical) on all 64); the section is bilateral by construction – on a bilateral box $\mathrm{Im}\, R$ changes sign, so a one-sided constructor with a fixed σ

cannot exist, and the negative control that forces one is refused 64/64; the target chart gauge is bounded away from zero; recentring is anisotropic ($2H$ in the imaginary direction, H in the real one) by an exact affine substitution in rational arithmetic, adding no remainder; and the bridge sign pattern is derived ($(1, -1, -1)$ on all 64), as in Section 6.1.

Two senses of "exact" occur in this section and are kept apart. An *algebraic* identity – the cocycle law, the deck relation of Proposition 7.1 – is derived symbolically and holds identically. A *certified residual* is a number with an enclosure: when a regluing is reported at 4.48×10^{-15}, that is an outward-rounded bound on a difference, not a proof that the difference vanishes. Both are certified; only the first is an equality.

Each bridge recloses on the lower side with a certified residual: open overlap of certified minimal width 5.49×10^{-4} in all four coordinates, anchor strictly interior, sheet separation at least 1.042, recentred difference 4.48×10^{-15}.

7.3 Continuation versus conjugation

Proposition 7.1 (the deck discrepancy). On the upper side of the bridge panel, the analytically continued section does *not* reglue to the conjugated atlas: the conjugate sections fail the exact regluing test on all 64 bridges. It reglues exactly to the *deck-translated* conjugate $D \cdot Z_{\mathrm{conj}}$, with $D = \mathrm{diag}(+1, -1, +1, +1, +1, -1)$, on all 64 bridges, with $O(1)$ margins. The identity is discriminating: changing any single sign of D breaks it (negative control).

In slogan form: the *derived mirror chart* and the *analytically continued sheet* are different objects, and their exact discrepancy is the deck transformation D (Figure 4). Conjugation of the chart data is a legitimate construction – but it is antiholomorphic in nature and it lands on the other sheet, so the conjugated description is a *diagnostic*, related to the continued one by D.

The sign pattern has a predictable geometric reason. On the real corner $\{\mathrm{Im}\, u = \mathrm{Im}\, v = 0\}$ the radicand R is real: where $\mathrm{Re}\, R > 0$ the principal root is real and the antiholomorphic involution *fixes* it; where $\mathrm{Re}\, R < 0$ the canonical root $i\sqrt{-R}$ is purely imaginary and the involution *negates* it. The mixed sign pattern of D is exactly the record of which solved lines sit in which regime.

7.4 Exact overlap closure and the certified nerve

Beyond the bridges, the panel closes into a nerve in which *an edge is a certified transition*, never a mere geometric intersection: 380 nodes (316 lower charts + 64 bridges); 5396 lower-lower edges imported from the prior panel and re-verified; 64 bridge-lower edges certified; 210/210 bridge-bridge edges certified (minimal margin 1.043, sup difference 7.94×10^{-15}); 588 new triples; the nerve is connected. The purely geometric intersection graph (338 edges) is published separately under its true name and carries no analytic claim.

Continuation also crosses codimension-one real faces: on a witness tile the continued section traverses the Re-face into the neighbouring canonical cell, and the sheet reached is *identified by a control experiment, not assumed*: neither recorded label closes naively; running the same experiment with the home label on the home side reproduces the exact predicted mismatch pattern $\theta = \varepsilon_{\mathrm{derived}} \odot \mathrm{label}$, elucidating the discrepancy as a measured convention difference between the register convention and the sign-certified regime convention. The identification is therefore *relative* and exact.

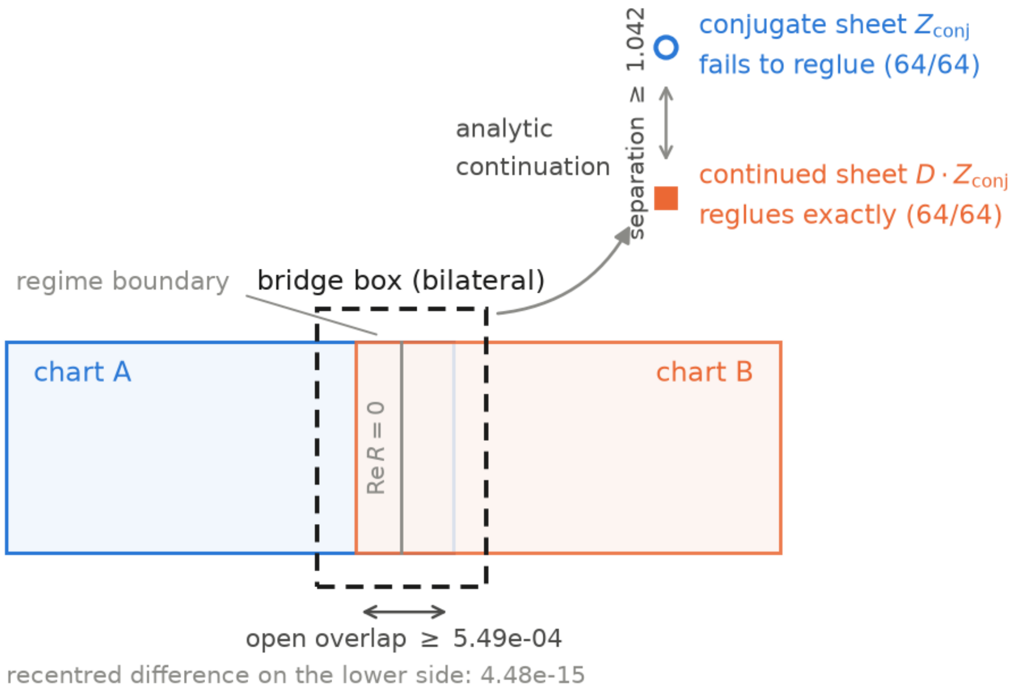

Figure 4: Continuation across a certified bridge: open overlap, regime boundary, and the conjugate sheet versus the continued sheet $D \cdot Z_{\mathrm{conj}}$. *(schematic, annotated with certified margins)*

The body exhibits three worked instances – one bridge with its certified margins, one face crossing with its control experiment, and the deck identity of Proposition 7.1 – each with the exact numbers quoted above. The full 64-bridge panel, the 380-node nerve with its 5,670 certified edges, and every box enumeration stay in the supplement and the artifact record.

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8. Certification Architecture and Reproducibility

8.1 Proof layers

The layering is the architecture of the paper (Figure 5): exact algebra supplies the identities, the quantitative lemma turns them into a chart with a radius, certification discharges the hypotheses, and only then does the atlas close.

Claim	Proof mode
X is K3	exact algebra
branched-cover structure	exact algebra
transition formulas	exact symbolic identities
chart applicability	interval / directed arithmetic
quantitative radius	exact bound + certified inputs
branch regime on a box	interval sign certificate

Claim	Proof mode
overlap continuation	exact identity after certified applicability
exploratory metric probes	validated numerics, not theorem input

8.2 Negative controls

Each principal certificate is accompanied by at least one perturbation known to invalidate the relevant hypothesis (wrong pivot, mutated sign pattern, wrong branch, wrong transition, box crossing a branch locus), verifying that the test is discriminating rather than vacuous.

8.3 Reproducibility package

Exact equations; certificate JSONs; scripts; environment pins; SHA-256 hashes. Verification is a single command,

```
python3 verification/verify.py
```

which distinguishes three levels, and the distinction is the point. The cheap certificates (each under a second) are **re-executed** and checked for all checks green, all negative controls green, and an unchanged outcome; a regenerated artifact is deliberately *not* hash-compared, since its provenance field changes with every commit and comparing it would test the repository rather than the mathematics. The coverage enumeration is **recomputed** and its counters compared one by one with the shipped file, gauge by gauge: it is the one certificate whose verdict would otherwise merely be read back. The expensive certificates – the bridge panel, the metric path, the face crossing – are **hash-verified** against recorded SHA-256 values; reproducing them is a matter of hours on a compute machine rather than of this script, and everything that run needs ships with the repository (Appendix F.4). The verifier fails, with a nonzero exit code, on any altered hash or any red condition, and it is read-only on the working tree: replaying rewrites provenance fields, so the original bytes are snapshotted and restored. Large box-enumeration artifacts never enter the PDF: dataset/supplement only.

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9. Validated Metric Compatibility on Selected Continuation Domains

As a downstream demonstration that the certified analytic atlas can support validated tensorial computations, we report certified relative residuals of order 10^{-9} on bridge/edge domains and a positive Weyl margin, under a preregistered threshold $\delta = 10^{-5}$. The panel runs on one certified cell ($S = \{0, 1, 5\}$, gauge z_2, sign pattern $\varepsilon = (1, 1, 1)$) against a threshold **fixed before the run**, $\delta = 10^{-5}$. On the 64 bridge charts the relative residual of the metric path is at most 5.93×10^{-9}; on the 274 certified nerve edges it is at most 3.19×10^{-9}; the Weyl slack stays above 2.98×10^{-2} throughout, and the frozen target-kind pattern (principal, rotated, principal) holds on 64/64.

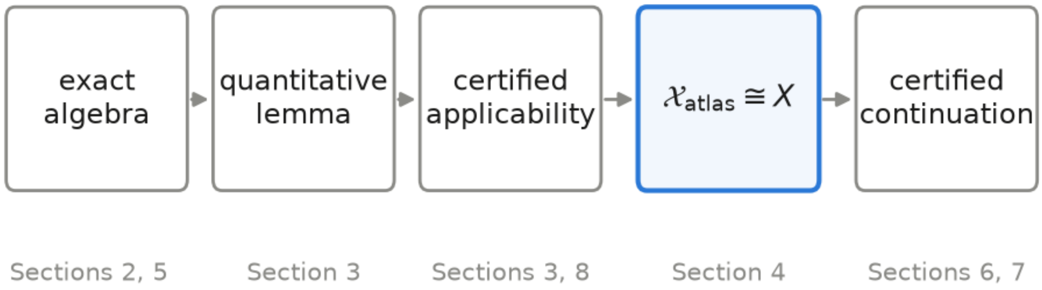

Figure 5: Exact algebra → quantitative lemma → certified applicability → $\mathcal{X}_{\text{atlas}} \cong X$ → certified continuation. *(schematic)*

Two honest qualifications. First, metric congruence is certified *below* δ, not exactly: these are enclosures of a residual, not an identity. Second, the certificate runs the metric path from one representative on each certified domain; **no independent two-sided metric comparison is claimed**, and the artifact records that fact explicitly.

This section does not establish a Ricci-flat or hyperkähler metric on X.

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10. Relation to Existing Explicit and Numerical K3 Constructions

10.1 Algebraic explicitness

Quartic, CI(2,2,2), Kummer and elliptic K3 models are classical, and the general theory of these surfaces is standard [1, 2]. The novelty here is not the existence of the equations.

10.2 Analytic and asymptotic constructions

Yau's theorem [3] gives existence of the Ricci-flat Kähler metric without any explicit description; Gross-Wilson [4] and the Kummer gluing tradition, up to recent work [5], prove existence and asymptotics by analytic means. Our object is not a new abstract existence proof, and this paper claims no metric at all.

10.3 Numerical K3 geometry

Donaldson's balanced-metric algorithm [6], the Headrick-Wiseman K3 computations [7], the energy-functional approach of Headrick and Nassar [8] – applied by them to the Fermat quartic K3 – the projective-embedding methods of Douglas-Karp-Lukic-Reinbacher [9] and the machine-learning approximations of the `cymetric` line [10] produce global numerical fits of Calabi-Yau metrics. The cited implementations do not provide interval-enclosed chart domains or exact transition certificates of the kind assembled here, and their reported residuals are diagnostics rather than enclosures. The two lines of work answer different questions.

10.4 Computer-assisted proofs in complex geometry

The closest methodological neighbour is recent: Ishige and Takayasu [11] compute the monodromy of Picard-Fuchs equations for a family of K3 toric hypersurfaces by rigorous analytic continuation along contours, controlling truncation and rounding with interval arithmetic. Their certified continuation lives in the *base parameter of a family*; ours lives in *charts on a fixed surface*, and the two are complementary. The general validated-numerics toolkit we rely on is standard [12, 13, 14, 15], as is the Newton-Kantorovich/Krawczyk lineage behind Proposition 3.2 [16, 17].

Against that background the claim of this paper is deliberately narrow:

> To our knowledge, the combination of an explicit CI(2,2,2) model, quantitative certified charts with a guaranteed radius, exact branch-aware transition data, and reproducible computer-assisted continuation between charts has not previously been provided for a concrete K3 surface.

We do *not* claim priority for certified continuation on K3 data in general [11], and we make no claim about computer-assisted proofs of Calabi-Yau metrics, which are the subject of separate ongoing work.

11. Discussion

11.1 What has actually been achieved

exact algebraic model → quantitative local charts → certified transitions → global analytic closure.

11.2 What remains open

No certified global Ricci-flat metric; no exact hyperkähler identity; no K3 family with controlled constants; no seven-dimensional gluing result. (The internal sign pattern separates the closed fixed-K3 atlas theorem from the open metric and family programmes.)

11.3 Why this infrastructure matters

The atlas is designed as an analytic substrate for validated PDE, metric and deformation computations on the same explicit surface.

Use of large language models

This work was carried out in sustained iteration with two large language models: Claude (Anthropic) and GPT (OpenAI). Their use went well beyond assisted copy editing, and it is therefore documented here rather than left undeclared. The certificate scripts, the negative controls, the derivations reported in the appendices and this manuscript were developed over repeated cycles of drafting, criticism and revision between the author and the two models; the models also assisted in checking references and code outputs against their sources. Two defects in the certificates were found in exactly this way, while writing the argument out in continuous prose for an outside reader, and both are recorded in the paper: an invariance checked under one group generator instead of the whole group, and a bound on a product used where a bound on a minimum was required. Neither model is an author: authorship carries accountability, and every condition, every run and every sentence of this paper was reviewed, verified and decided by the author, who takes sole responsibility for its accuracy, integrity and originality.

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Statements and Declarations

Competing interests. The author has no financial interests, and no institutional or funding relationship, bearing on this work.

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Data and code availability. Every quantitative claim in this paper is backed by a machine-checkable certificate. The certificate scripts, their JSON outputs, the producers of the expensive certificates with the inputs they read, the figure sources and the verification entry point of Appendix F are publicly available in the repository accompanying this paper (github.com/Arithmon/K3), and archived on Zenodo under the concept DOI 10.5281/zenodo.22047469, which resolves to the latest deposited version. Verification is a single command, and it distinguishes three regimes: the certificates it re-executes, the coverage enumeration it recomputes and compares counter by counter, and those it verifies by hash (Appendix F).

Ethics approval. Not applicable. The work involves no human participants, no animal subjects and no personal data.

Author contributions. Sole author: the author designed the certificates, wrote the analysis code, ran the verifications and wrote the manuscript. Large language models were used throughout, in the manner and to the extent documented above; they meet no authorship criterion, and accountability for the work is the author's alone.

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Appendix A – Exact defining equations

A.1 The data. $\mu = (\mu_0, \ldots, \mu_5) = (1, 2, 3, 5, 7, 11)$ and $F_k(z) = \sum_{j=0}^{5} \mu_j^k z_j^2$ for $k = 0, 1, 2$; $X = \{F_0 = F_1 = F_2 = 0\} \subset \mathbf{P}^5$. Everything else in the paper is a rational function of these six integers.

A.2 Vandermonde determinants. For a triple S, $V_S = \det(\mu_s^k)_{k=0,1,2;\, s\in S} \in \mathbf{Z}$. All twenty are nonzero (this is Lemma 2.1's ingredient); their absolute values are

S	$\lvert V_S\rvert$	S	$\lvert V_S\rvert$	S	$\lvert V_S\rvert$	S	$\lvert V_S\rvert$
012	2	013	12	014	30	015	90
023	16	024	48	025	160	034	48
035	240	045	240	123	6	124	20
125	72	134	30	135	162	145	180
234	16	235	96	245	128	345	48

with extremes $\min |V_S| = 2$ (at $S = \{0, 1, 2\}$) and $\max |V_S| = 240$. The largest transition coefficient over all triples is $C = \max_S \|V_S^{-1} V_T\|_\infty = 112$, attained at $S = \{0, 1, 2\}$; the largest single entry is 80.

A.3 Normalization. $c_k^2 = \sum_j \mu_j^{2k}$ gives $c_0^2 = 6$, $c_1^2 = 209$, $c_2^2 = 17765$. The squares are integers, the c_k are not. The **algebraic layer** (Sections 2, 4, 5 and Appendices A, C) lives over $\mathbf{Q}$ with $V_S \in M_3(\mathbf{Z})$; the **normalized layer** (Section 3 and Appendix B) is a conditioning convention, and the certificate brackets the c_k by outward-rounded rationals rather than approximating them.

A.4 Distinguished patch. $S = \{3, 4, 5\}$, gauge $z_0 = 1$, base $(u, v) = (z_1, z_2)$: the radicands and the degree-six discriminant are displayed in Section 2.3, and the corresponding regime geometry is Figure 1.

Appendix B – Proof of Proposition 3.2

B.1 Setup. Fix a chart type (S, g). Write $w = z_S \in \mathbf{C}^3$ for the solved variables, $b = (u, v) \in \mathbf{C}^2$ for the base, $z_g = 1$ for the gauge, and $G(w; b) := \tilde{F}(w, b, 1) : \mathbf{C}^3 \times \mathbf{C}^2 \to \mathbf{C}^3$ for the normalized equations. Let w_0 solve $G(w_0; b_0) = 0$ at a centre b_0 in the sector, so $|b_0| \le 1$.

B.2 Three exact identities. Because the F_k are diagonal quadrics, the following hold identically – they are not estimates:

- **(E1)** $\partial_w G(w; b) = \tilde{M}(w) = 2\,\tilde{V}_S\,\mathrm{diag}(w)$, independent of b;
- **(E2)** $G(w; b) - G(w_0; b) - \tilde{M}(w_0)(w - w_0) = \tilde{V}_S \cdot ((w - w_0)^2)$, the square taken componentwise: the Taylor remainder is one explicit quadratic term and the "Hessian" is constant;
- **(E3)** $G(w_0; b) - G(w_0; b_0) = \tilde{V}_{uv} \cdot (b^2 - b_0^2)$.

B.3 Residual. Let $\sigma \le s_{\min}(\tilde{M}(w_0))$ and let $a_{S,g}$ bound $\|\tilde{V}_{uv}\|$. For $|b - b_0| \le \rho$, (E3) and $G(w_0; b_0) = 0$ give $\|G(w_0; b)\| \le a_{S,g}\,\rho\,(2|b_0| + \rho) \le a_{S,g}\,\rho\,(2 + \rho)$, hence the simplified-Newton residual

$$\eta \;:=\; \|\tilde{M}(w_0)^{-1} G(w_0; b)\| \;\le\; \frac{a_{S,g}\,\rho\,(2+\rho)}{\sigma}.$$

B.4 Contraction. Let $T(w) = w - \tilde{M}(w_0)^{-1}G(w; b)$. By (E1),

$$DT(w) = I - \tilde{M}(w_0)^{-1}\tilde{M}(w) = 2\,\tilde{M}(w_0)^{-1}\tilde{V}_S\,\mathrm{diag}(w_0 - w),$$

so $\|DT(w)\| \le (L/\sigma)\|w - w_0\|$ with $L = 2\,\sigma_{\max}(\tilde{V}_S)$. With

$$h \;=\; \frac{L\,\eta}{\sigma} \;<\; \frac{1}{2},$$

T maps the closed ball of radius $r^* = (1 - \sqrt{1-2h}\,)\,\sigma/L$ around w_0 into itself and contracts there; $r^* \le 2\eta$. Hence a zero $w(b)$ exists in that ball, and it is unique in it.

B.5 Uniqueness in the large ball. For diagonal quadrics the midpoint identity is exact: for any w, w',

$$G(w; b) - G(w'; b) \;=\; \tilde{M}\left(\frac{w + w'}{2}\right)(w - w'),$$

since $w_s^2 - w_s'^2 = 2 \cdot \frac{w_s + w_s'}{2} \cdot (w_s - w_s')$ term by term. If w, w' are two zeros with $\|w - w_0\| < \sigma/L$ and $\|w' - w_0\| < \sigma/L$ (open ball, so the perturbation estimate stays strictly below the invertibility threshold), then $\tilde{M}((w + w')/2)$ is invertible there and $w = w'$. Uniqueness therefore holds on a ball much larger than r^*.

B.6 Holomorphy. Start the iteration at the constant w_0. Since G is quadratic and $\tilde{M}(w_0)$ is a fixed matrix, each iterate $T^n(w_0)$ is a *polynomial* in the base variables b. The convergence of B.4 is uniform on the ball, so the limit $w(b)$ is holomorphic in b, and the chart map $b \mapsto (w(b), b)$ is a holomorphic section.

B.7 Certified constants. All quantities above are bounded by exact rationals: $\sigma_{\max}(\tilde{V}_S) \le \|\tilde{V}_S\|_F$ (upper), $\sigma_{\min}(\tilde{V}_S) \ge |\det \tilde{V}_S|/\|\tilde{V}_S\|_F^2$ (lower), square roots bracketed by integer square roots to 10^{-40}. The checks verify $h < 1/2$, the contraction factor, $r^* \le 2\eta$ and $r^* < \sigma/L$ on all 60 types, and a negative control multiplying ρ by 100 drives h above 1/2. The resulting uniform radius is $\rho_{\mathrm{unif}} = \min_{S,g} \rho_{S,g} \ge 2.10 \times 10^{-12}$, the per-type values spanning $[2.10 \times 10^{-12},\ 1.71 \times 10^{-4}]$ (Figure 2); these are Frobenius/determinant bounds, deliberately not sharp.

The lower bound on σ deserves its own line, because it is where an earlier version of this certificate was optimistic. Since $\tilde{M}(w_0) = 2\,\tilde{V}_S\,\mathrm{diag}(z_S)$,

$$s_{\min}(\tilde{M}(w_0)) \;\ge\; 2\,s_{\min}(\tilde{V}_S)\,\min_{s \in S} |z_s|,$$

so what is needed is the **minimum** of the solved coordinates. The pivot threshold bounds only their **product**; substituting one for the other is invalid whenever a solved coordinate exceeds 1, which happens on certified domains, and an explicit point of X on a certified chart domain witnesses the failure. The valid bound on $\min_s |z_s|$ is Lemma 3.1, and the radius quoted above is the one it yields.

Appendix C – Transition formulas

C.1 The square-level cocycle. The quadrics are linear in $w_j = z_j^2$, so $V_{\text{full}}\, w = 0$ determines all six squares from any admissible triple. For each ordered pair (S, S') this yields an exact rational matrix $P_{S'S} \in M_3(\mathbf{Q})$ with $w_{S'} = P_{S'S} w_S$ on X, obtained by composing $w_T = -V_T^{-1} V_S w_S$ with the selection of the S' entries. Verified in exact arithmetic: $P_{SS} = \mathrm{Id}$ (20/20), $P_{SS'} P_{S'S} = \mathrm{Id}$ (400/400), and $P_{S''S'} P_{S'S} = P_{S''S}$ on all $20^3 = 8000$ ordered triples.

C.2 Elementary swap. For $S = \{3,4,5\} \to S' = \{2,4,5\}$,

$$P_{S'S} = \begin{pmatrix} -6 & -15 & -45 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \text{i.e.} \qquad w_2 = -6w_3 - 15w_4 - 45w_5.$$

Exactly one row differs from the identity – the row of the newly solved coordinate. On coordinates the map is $z_{s'} = \varepsilon_{s'} \sqrt{(P_{S'S} w_S)_{s'}}$, so the sign pattern travels with the transition.

C.3 Gauge change. $(S, g) \to (S, g')$ is the projective rescaling by $z_g / z_{g'}$; it changes no sign pattern and no solved triple.

C.4 Generation. The twenty triples with adjacency $|S \cap S'| = 2$ form $J(6,3)$: 20 nodes, 90 edges, connected, diameter 3. The 60 chart types under {swap, gauge change} form a connected graph of diameter 4 – the swap holds g fixed and the gauge change holds S fixed, so a transition whose target triple contains the source gauge needs a gauge move of its own. With C.1, every transition is a word of length at most 4 in these two moves composed with a single sheet-group element (Proposition 5.1). Removing gauge changes disconnects the type graph into six components of ten.

C.5 Scope. C.1–C.4 are algebraic identities over $\mathbf{Q}$. That a given transition *applies* on a given domain – gauge nonvanishing, branch regime constant, overlap open – is a separate, certified statement (Sections 7 and 8).

Appendix D – Branch conventions

D.1 Chart data. A chart carries: a solved triple S; a gauge $g \notin S$ with $z_g = 1$; base variables $b = (u, v)$; and a sign pattern $\varepsilon \in \{\pm 1\}^3$ with $z_s = \varepsilon_s \sqrt{R_s(b)}$. The sector $C_{S,g} = \{|z_g| = \max_{t \in T} |z_t|\}$ is a **selector** of representatives (it gives $|u|, |v| \leq 1$ and feeds Lemma 3.1); it is never used as an open set of a cover.

D.2 Determinations. $\sqrt{\cdot}$ is the principal square root, cut $(-\infty, 0]$. The **principal** determination of $\sqrt{R}$ is available where R avoids $(-\infty, 0]$; the **rotated** (canonical) determination $\sqrt{R} = \sigma\, i \sqrt{-R}$ is available where R avoids $[0, +\infty)$, the two conditions being exchanged by $R \mapsto -R$. The sign is not free:

$\arg R$	$\arg(-R)$	$i\sqrt{-R}$	σ
$(0, \pi]$	$\arg R - \pi$	$+\sqrt{R}$	$+1$
$(-\pi, 0)$	$\arg R + \pi$	$-\sqrt{R}$	-1

On a bilateral box $\operatorname{Im} R$ changes sign, so no single σ exists there – which is why bridge sections are built bilaterally (Section 7.2), and why forcing a component σ on a bridge is refused 64/64.

D.3 Sign groups. Diagonal sign matrices preserve every F_k. Modulo the projective scalar -1 they form the global group $G_{\text{sign}} \simeq (\mathbf{Z}/2)^5$ of order 32, generated by $\sigma_1, \dots, \sigma_5$ with $\sigma_0 = -\sigma_1\sigma_2\sigma_3\sigma_4\sigma_5$. For a fixed projection π_S the deck group is the vertical subgroup $G_{\text{deck}}(S) \simeq (\mathbf{Z}/2)^3$, of order 8 and index 4: an element is vertical for (S, g) iff its sign vector is constant on the complement T. For $D = \operatorname{diag}(+1, -1, +1, +1, +1, -1)$ this holds for exactly 4 triples, i.e. 12 of the 60 types (Figure 3).

D.4 Continuation versus conjugation. On the real corner $\{\operatorname{Im} u = \operatorname{Im} v = 0\}$ the radicand is real, and the antiholomorphic involution acts on each solved line according to its regime:

regime on the line	root	conjugation acts by
$\operatorname{Re} R > 0$ (principal)	real	fixes it
$\operatorname{Re} R < 0$ (canonical)	purely imaginary $i\sqrt{-R}$	negates it

The mixed sign pattern of D is exactly the record of which lines sit in which regime; this is the geometric content of Proposition 7.1.

D.5 Discipline. Ledgers are **derived, never defaulted**: in the bridge construction the default $(1, 1, 1)$ was refused by the exact regluing test and the derived $(1, -1, -1)$ closed it. Sheet statements are made relative to a chart's sign pattern convention, never as absolute class labels; identifications across conventions are measured, as in the face-crossing control experiment $\theta = \varepsilon_{\text{derived}} \odot \text{label}$ (Section 7.4).

Appendix E – Certificate specification

E.1 Anatomy. A certificate is one JSON document. The vocabulary is shared, but it is not a schema every file satisfies, and stating otherwise would be a claim the reader can falsify in one command. Counted as TOP-LEVEL fields over the fourteen shipped files – the level at which a skeleton is a skeleton, and not "the key appears somewhere in the document":

field	certificates carrying it
`artifact`	13
`checks`, `checks_passed`, `checks_total`	12
`built_from_head`, `self_sha256`	10
`kind`, `seconds`	9
`upstream`, `perturbation_tests`	8
`does_not_attest`, `subject`	6
`outcome`, `provenance`	5
`n_atteste_pas`	4

Three of the files nest `self_sha256` inside a `provenance` block instead of carrying it at the top level, which is why that row reads 10 and not 13.

The fields mean: an `artifact` name; a `subject` line stating the claim; a `kind` (exact algebra, interval, closed form); `upstream`, holding the SHA-256 of every certificate consumed; the payload blocks; `checks`, a dictionary of **named booleans**; `checks_passed / checks_total`; `perturbation_tests`, the negative controls; an `outcome` string; the explicit list of what the certificate does *not* establish – written `does_not_attest` in the six most recent files and `n_atteste_pas` in the four oldest, a rename this repository did not finish; and the provenance triple `seconds`, `built_from_head`, `self_sha256`.

The four design records are the least uniform: they predate the convention – none of them carries `does_not_attest`, `outcome`, `subject` – but all four carry `checks` and its two counters (6/6, 4/4, 7/7, 6/6). Of the nine files the verification command covers, 8 carry `checks`; the ninth, the recomputed coverage certificate, has no `checks` block at all and is compared field by field on its `verdict` and its box counts instead (Appendix F).

E.2 Green and red. A certificate is green iff every condition and every negative control is true. The `outcome` string encodes the claim when green and collapses to a `*_checks_red` marker otherwise, so a red certificate cannot be quoted as if it were green.

E.3 Negative controls. Each principal condition ships with at least one deliberate mutation known to invalidate its hypothesis – a repeated μ, a wrong pivot threshold, a mutated sign pattern, a wrong branch, a falsified transition entry, a box crossing a branch locus. A condition that no mutation can redden is vacuous and is rejected rather than reported.

E.4 Derivation discipline. A separate lint enforces three rules on the certificates: constants that can be derived must be derived rather than named (D1); a certificate must not silently consume another block's output (D2, blocking); and no condition may be constant-true by construction (D3).

E.5 Claim boundary. `n_atteste_pas` is not a disclaimer but part of the specification: it is where the metric probe of Section 9 records that no independent two-sided comparison was run, and where the atlas certificates record that the quantitative floor, not the covering, is what the enumeration audit supplies.

Appendix F – Reproduction instructions

F.1 One command.

```
python3 verification/verify.py
```

It prints one line per *checked* certificate and exits nonzero on any failure; a `--quick` flag checks hashes only, skipping both replay and recomputation. It is read-only on the working tree, and takes about two minutes, almost all of it in the recomputed coverage certificate.

The repository ships fourteen certificate files and the command checks nine of them: five replayed, one recomputed, three hash-verified (F.3). The other five are records rather than claims – four design documents (the uniform chart lemma, the exact transitions, the regional gluing contract, the closure skeleton) and the figure manifest. They do have checks of their own, and an earlier revision of this verifier replayed them; that was reverted on purpose. Shipping their producers would pull twenty-three further artifacts into the repository through what those producers read – contract amendments, preregistrations, an internal chain none of the theorems above rest on – and a repository that carries what it does not need is harder to audit, not easier.

F.2 Environment. The reference environment is pinned, not recommended: Python 3.14.4, NumPy 2.5.1, SciPy 1.18.0, SymPy 1.14.0, Matplotlib 3.11.1 and mpmath 1.3.0, as listed in `requirements.lock` and `pyproject.toml` and materialised by the `Dockerfile`; `docs/ENVIRONMENT.md` records the BLAS backend, the determinism variables and the platform each certificate was produced on. The verification command itself imports only NumPy, SymPy, Matplotlib and mpmath, and has also been run outside the pinned environment, under Python 3.12 with mpmath 1.3.0; the package install (`pip install -e .`) requires Python 3.14.4 exactly. The mpmath pin is enforced by `verify.py` itself, which checks the installed mpmath before anything else runs and refuses to continue against a different version, because that version is serialized in the provenance of the whole chain.

F.3 Three tiers. Five cheap certificates (each under a second) are **re-executed** and checked for all checks green, all negative controls green and an unchanged outcome: the chart theorem, the atlas closeout, the generator/obligation certificate, the smoothness/coverage/transition certificate, and the sigma-floor correction whose repair is reported in Appendix B.7 – that last one is replayed precisely so that the correction is checked by the reader rather than asserted by the author.

The coverage certificate of Section 4.1 is **recomputed**: its branch-and-bound enumeration is re-run and its counters are compared one by one with the shipped file, gauge by gauge. This is the exhaustive enumeration behind the observed minor floor 4.8 – 71,807,792 boxes in float64 interval arithmetic with one-ulp outward rounding, about seventy seconds on four cores – and it is the computation on which the preregistered threshold $m_{\text{alg}} = 4$ has its margin. Reading its verdict would test nothing; re-running it reddens if a single counter moves. The published run is reproduced exactly, box for box, on all six gauges.

The three expensive ones are **hash-verified**:

artifact	SHA-256 (first 16)
bridge panel (64 charts, nerve)	`58c06da48412c8b1`
metric path, full (64 bridges, 274 edges)	`397c19b105819f79`
face crossing (control experiment)	`9088e59844c7e0dd`

F.4 What re-running does not test. A regenerated artifact is deliberately not hash-compared: its provenance field changes with every commit, so such a comparison would test the repository rather than the mathematics. Reproducing the expensive certificates from scratch is not part of this script, but nothing that run needs is withheld: the producers of the bridge panel, the metric path and the face crossing ship as the package `src/k3_atlas`, the five shared numerical kernels included, together with the 25 input artifacts they read (`src/k3_atlas/data`, 41 MB), the pinned environment of F.2 and a `Dockerfile` that reproduces it. For each result of the paper, `docs/RESULTS_INDEX.md` names the generator, the reference artifact and the reproduction mode that rebuilds it, and gives the measured cost of the full chain. Each generator is an ordinary Python module, run in the pinned environment; the full chain takes hours, which is why it is kept apart from the one-command check. The figures are regenerated by their own deterministic script, whose manifest records every plotted number and the hashes of its sources.